How wide is rue de Terrebonne? The roadway along the lane, from the City's 2015 LiDAR

Terrebonne-open-sources, 2026-11-04. Data: Ville de Montréal, LiDAR aérien 2015 and unités d'évaluation foncière (open data, CC-BY 4.0). Method and code: `https://openmobilitydata.github.io/Terrebonne-open-sources/2015-lidar-terrebonne-street-width/`.

Question

The winter layout proposed for rue de Terrebonne (borough file 1266880005, adopted November 2, 2026) keeps the contraflow protected lane and turns the lane running with traffic into parking for the winter, so that cyclists going with traffic share the traffic lane. The cross-section drawn for that type of layout in the borough's 2023 study (EXP, Figure 6-6, p. 72) is, from the south property line: sidewalk 1.8 m, contraflow lane 2.0 m, buffer 0.5 m, parking 2.4 m, shared lane 3.5 m, parking 2.4 m, sidewalk 1.8 m, that is 10.8 m from curb to curb in a 14.4 m section. Whether the street is in fact 10.8 m wide, everywhere along the 2.4 km, decides how much room any winter layout has. This note measures it.

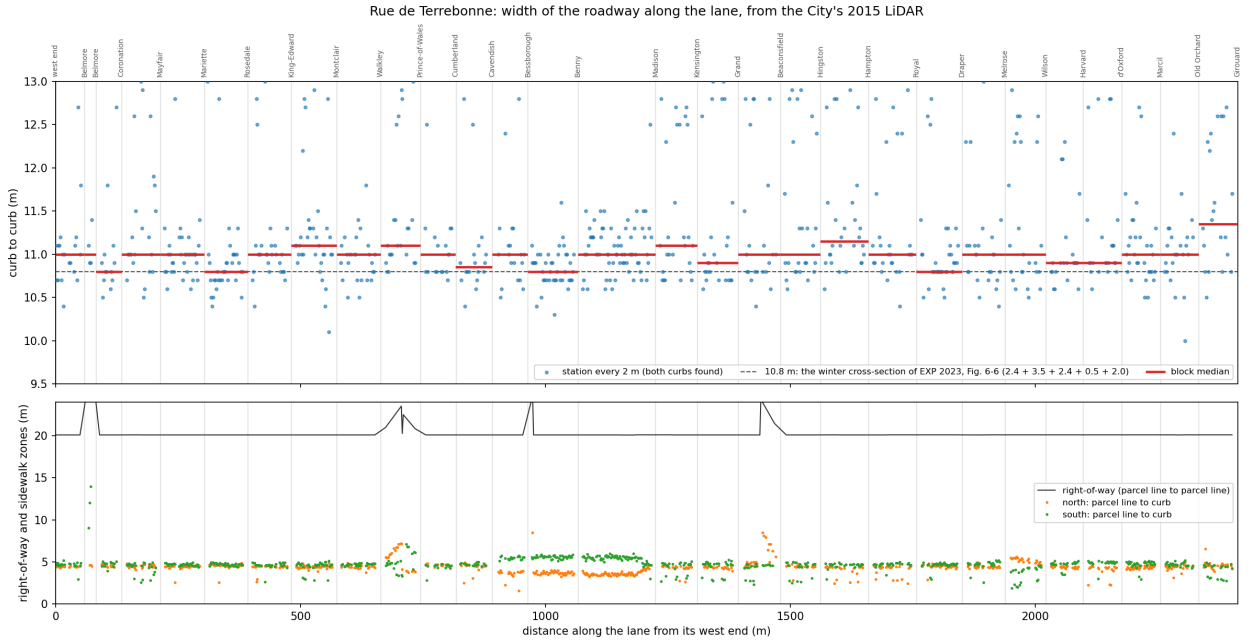
Data and method

The City's 2015 airborne LiDAR (six 1 km tiles, NAD83(CSRS)/MTM 8; classes ground, high vegetation, building) was rasterized once for the earlier parking work: the mean elevation of ground-class returns on a 0.25 m grid along the corridor. On that surface a cross-section is sampled every 2 m along the lane's axis (the City géobase, from the lane's west end to Girouard; 1207 stations), from 16 m south to 16 m north of the axis, every 0.1 m. On each side, scanning outward from the axis, the curb is the first place between 3 and 8 m from the axis where the ground rises by at least 7 cm over 0.5 m; its position is the point of steepest rise in that window. A Montréal curb face is 10 to 15 cm high; a driveway ramp at about 1:12 rises 4 cm over 0.5 m and does not pass, though a steeper apron sometimes does. The LiDAR was flown on two weeknights with cars parked at the curbs, and under a car or a dense tree there are no ground returns; a station is kept only when both curbs are found and the width lies between 9 and 13 m (759 stations). The right-of-way is measured on the same line as the distance between the first parcel outline on each side (the City's assessment units); the sidewalk zone is the parcel line to the curb.

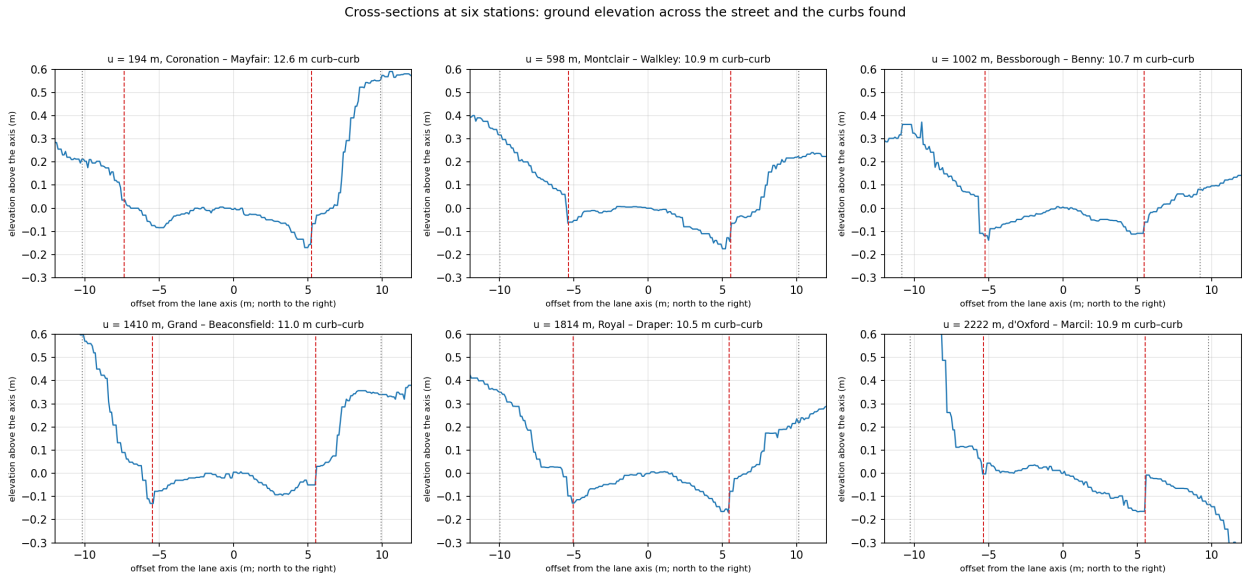
Results

- Curb to curb: median 11.0 m over the 759 stations kept; 10 % of stations read under 10.7 m and 10 % over 12.4 m (the high tail is aprons taken for curbs). The block medians run from 10.8 to 11.35 m: the roadway is the same width along the whole lane, within a few decimetres.
- Right-of-way: 20.09 m at every block but the Belmore stub and three lots (the 66-foot standard, 20.12 m); sidewalk zones about 4.4 m on the north side and 4.73 m on the south side (parcel line to curb, including the grass strip).
- EXP's own statement of the existing section (report p. 50, PDF p. 61): « L'emprise fonctionnelle actuelle de la rue de Terrebonne est d'environ 14,6 m, soit une chaussée d'environ 11 m et deux trottoirs d'environ 1,8 m. » The LiDAR's 11.0 m is EXP's « environ 11 m »; EXP's 14.6 m is roadway plus paved sidewalks, not the 20.1 m right-of-way, the difference being the grass strips.

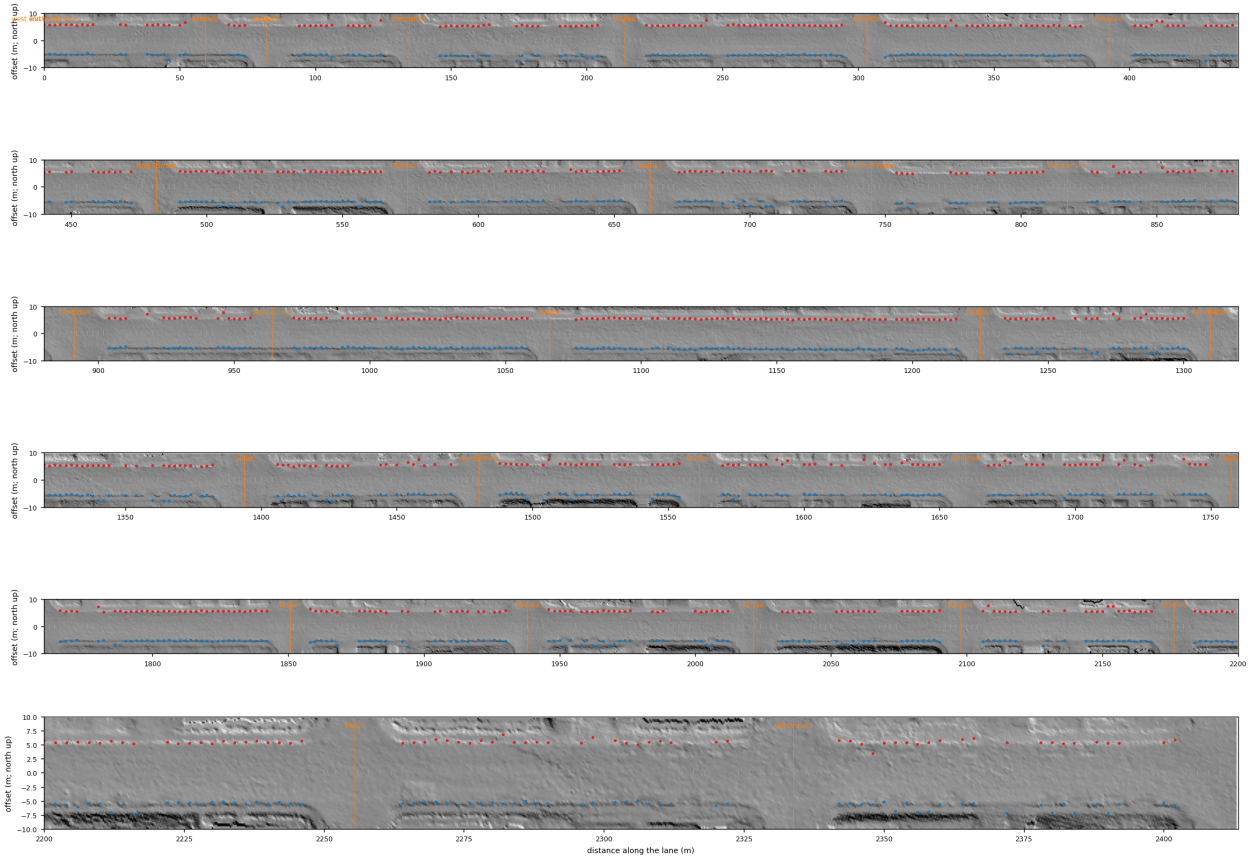
- Against the 2023 drawing: 10.8 m of lanes and parking in an 11.0 m roadway. The winter layout fits with about 0.2 m to spare across the street. A shared lane wider than 3.5 m would have to come from the parking bands (2.4 m), the buffer (0.5 m) or the contraflow lane (2.0 m).
- Before 2024 the street was two-way with parking on both sides; two 3.3 m lanes (EXP 2023) and two parking bands of 2.2 m make the same 11.0 m.



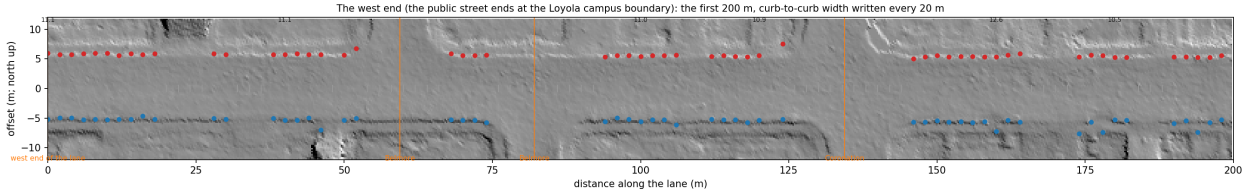
Ground elevation: Ville de Montréal, LiDAR aérien 2015 (CC-BY 4.0), ground returns averaged on a 0.25 m grid; parcels: Ville de Montréal, unités d'évaluation foncière (CC-BY 4.0). Curb = the first rise of 7 cm over 0.5 m outward from the axis, between 3 and 8 m. Stations where a parked car or a tree hid a curb are left out.



Review figure: the LiDAR ground surface in the street frame (hillshade, 0.25 m) with every curb detection; red = north curb, blue = south curb, grey ticks on the axis = station dropped (no ground under a parked car or a tree, or a width outside 9–13 m)



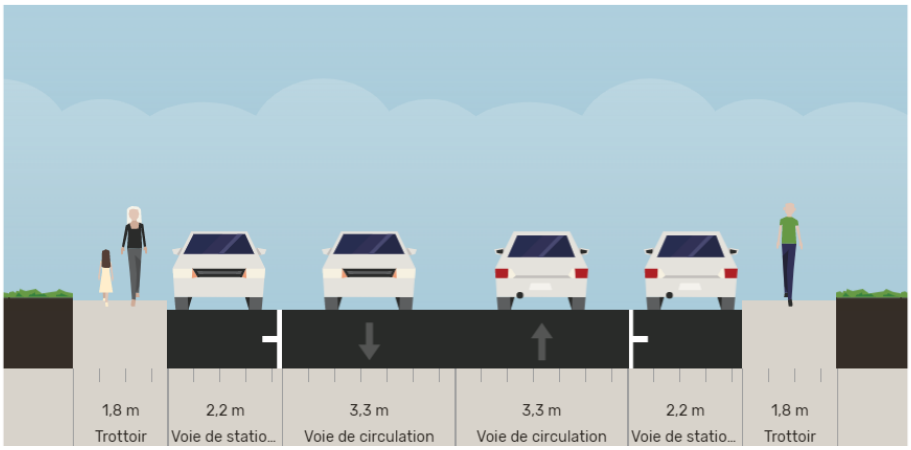
The west end (the public street ends at the Loyola campus boundary): the first 200 m, curb-to-curb width written every 20 m



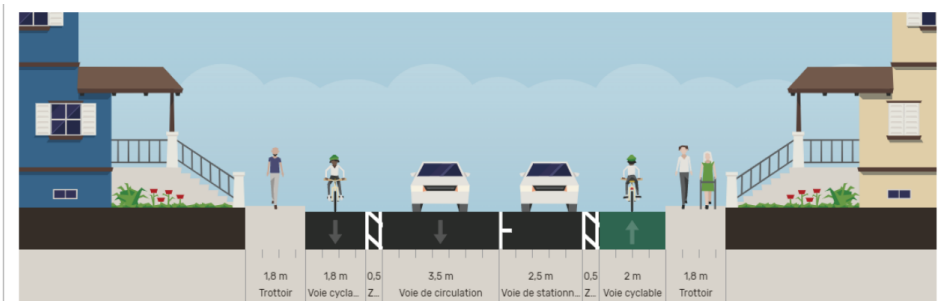
The west end

The public street ends where the lane ends, at the Loyola campus boundary (the City géobase's last segment, 1320365, ends there; beyond it the roadway is the campus's). The last public block, west end to Belmore, reads 11.0 m curb to curb (19 of 30 stations kept), a 20.09 m right-of-way and sidewalk zones of 4.4 m north and 4.7 m south: the same section as the rest of the street. It is two-way with no parking on either side; the 2015 surface and the pre-2024 imagery cannot measure the 2024 bike lanes, but an 11.0 m roadway holds, for example, two 2.0 m lanes with 0.5 m buffers and two 3.0 m travel lanes (arithmetic, not the marking plan).

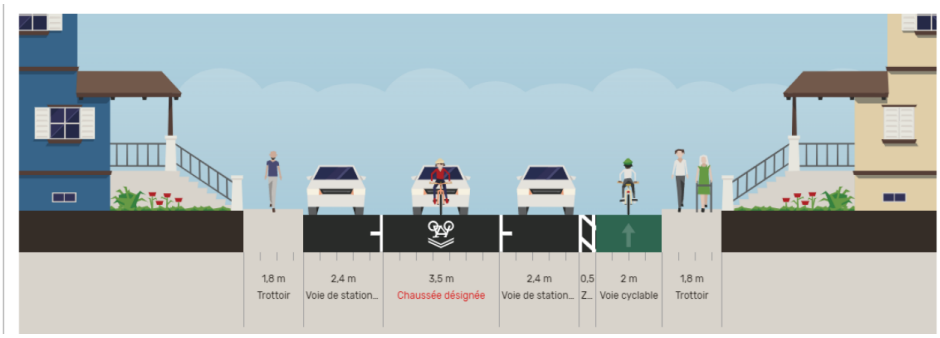
EXP's three sections against the measured roadway



2023 (EXP Fig. 3-1, report p. 25): 2.2 + 3.3 + 3.3 + 2.2 = 11.0 m curb to curb.



Today, Aménagement B (Fig. 6-7, p. 72): 1.8 + 0.5 + 3.5 + 2.5 + 0.5 + 2.0 = 10.8 m.



Winter, assumed to be Aménagement A (Fig. 6-6, p. 72): 2.4 + 3.5 + 2.4 + 0.5 + 2.0 = 10.8 m; applied to B's bands, 2.3 + 3.5 + 2.5 + 0.5 + 2.0 = 10.8 m.

Every section EXP drew fits the measured 11.0 m with 0 to 0.2 m to spare; none has room for a wider shared lane without narrowing the parking bands, the buffer or the contraflow lane. The figures are reproduced from the borough's published study (Terrebonne-open-sources, 2023-exp-mobility-study).

By block

Block (west to east)	kept/stations	curb–curb median (m)	p10–p90	right-of-way (m)	sidewalk N (m)	si
west end – Belmore	19/30	11.0	10.7–11.32	20.09	4.44	
Belmore – Belmore	4/12	11.0	10.9–11.31	28.48	4.6	

Belmore – Coronation	13/26	10.8	10.62–11.66	20.09	4.5
Coronation – Mayfair	23/39	11.0	10.72–12.6	20.09	4.51
Mayfair – Mariette	34/45	11.0	10.7–11.27	20.09	4.46
Mariette – Rosedale	34/45	10.8	10.53–11.24	20.09	4.59
Rosedale – King-Edward	27/44	11.0	10.76–11.78	20.09	4.46
King-Edward – Montclair	34/46	11.1	10.73–12.55	20.09	4.39
Montclair – Walkley	26/45	11.0	10.7–11.25	20.09	4.46
Walkley – Prince-of-Wales	22/41	11.1	10.9–12.78	21.94	5.54
Prince-of-Wales – Cumberland	17/36	11.0	10.8–11.3	20.09	4.67
Cumberland – Cavendish	18/37	10.85	10.67–11.59	20.09	4.53
Cavendish – Bessborough	19/37	11.0	10.6–11.52	20.09	3.75
Bessborough – Benny	41/51	10.8	10.7–11.1	20.09	3.69
Benny – Madison	68/79	11.0	10.7–11.43	20.09	3.56
Madison – Kensington	26/42	11.1	10.7–12.65	20.09	4.39
Kensington – Grand	25/42	10.9	10.7–12.8	20.09	4.38
Grand – Beaconsfield	22/44	11.0	10.71–12.8	20.09	4.98
Beaconsfield – Hingston	26/40	11.0	10.75–12.8	20.09	4.48
Hingston – Hampton	28/49	11.15	10.87–12.73	20.09	4.34
Hampton – Royal	27/49	11.0	10.8–12.52	20.1	4.38
Royal – Draper	35/47	10.8	10.68–11.36	20.1	4.48
Draper – Melrose	25/44	11.0	10.64–11.94	20.09	4.38
Melrose – Wilson	25/41	11.0	10.6–12.6	20.1	5.23
Wilson – Harvard	22/38	10.9	10.61–12.06	20.1	4.4
Harvard – d'Oxford	23/40	10.9	10.8–12.78	20.1	4.27
d'Oxford – Marcil	30/39	11.0	10.7–11.61	20.1	4.31
Marcil – Old Orchard	24/39	11.0	10.56–11.37	20.1	4.4
Old Orchard – Girouard	22/40	11.35	10.8–12.6	20.1	4.57

Caveats

- The curb is read from a 0.25 m elevation grid: a single reading is good to about 0.2 m; the block medians are the figures to use.
- 2015 surface, 2024 layout: the curbs have not moved; the lane markings of 2024 are not in the LiDAR.
- Stations hidden by parked cars or trees are left out, not interpolated; blocks with few kept stations (Hingston–Hampton, Hampton–Royal) carry more uncertainty.
- The right-of-way is the City's parcel geometry, not a survey.